

POWERFACTORY

PowerFactory 2024

Technical Reference

WECC Wind Turbine Templates

WECC WTG Type1 0.5MVA 50Hz,
WECC WTG Type1 0.5MVA 60Hz,
WECC WTG Type2 0.5MVA 50Hz,
WECC WTG Type2 0.5MVA 60Hz,
WECC WTG Type3 2.0MVA 50Hz,
WECC WTG Type3 2.0MVA 60Hz,
WECC WTG Type4A 2.0MVA 50Hz,
WECC WTG Type4A 2.0MVA 60Hz,
WECC WTG Type4B 2.0MVA 50Hz,
WECC WTG Type4B 2.0MVA 60Hz

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POWER SYSTEM SOLUTIONS
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1 Introduction

This document describes the usage of the WECC wind turbine generator (WTG) models in *PowerFactory*. The models are based on [1, 9].

The following four WTG types defined by WECC are available in *PowerFactory*:

- Type 1 (WTG with conventional induction generator)
- Type 2 (WTG with variable rotor-resistance induction generator)
- Type 3 (WTG with doubly fed induction generator (DFIG))
- Type 4A/B (WTG connected through full scale power converter)

Chapters 2 and 3 describe how to use and configure the models in *PowerFactory*. In chapter 4 the different models are described.

Note: The models which are available in *PowerFactory* as templates are configured with a set of sample parameters. The user has to adjust these parameters (i.e. power rating, voltage level, reactive power limits and all parameters of the dynamic models) to represent a specific WTG of a specific manufacturer!

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2 Model Usage

Each of the types consist of at least two grid elements, either the asynchronous generator (*ElmAsm*) or the static generator (*ElmGenstat*), a terminal and of a composite model which defines the dynamics of the different WTG types. The composite model contains the dynamic modelled using DSL (*DIgSILENT* Simulation Language) and all necessary measurement devices. The models are available as templates.

For each template generic protection devices are provided to act in case of over or under frequency and voltages during operation. The templates are accessible to be used in a project through the button in the drawing toolbox.

3 Model Configuration

The rated power, dispatched power and number of parallel machines can be adjusted in the templates considering the next changes.

3.1 Type 1 and 2

3.1.1 Adjusting Rated Power

The rated power is defined via the asynchronous generator type, which can be accessed via the asynchronous generator element on the *Basic Data* page. The rated power as well as other attributes such as the inertia of the generator can be changed in the type.

Note: Changing the type will affect all elements using this type. Create a copy of the type to avoid this.

3.1.2 Adjusting Active Power Dispatch

The dispatch of the generator is set on the *Load Flow* page. The default bus type is “AS” (for asynchronous generator). Note that by using this bus type, only the active power dispatch can be entered. The reactive power will be calculated via the machine parameters defined in the type. The reactive power demand will be compensated with the shunt capacitor included also within the Type 1 and Type 2 templates.

3.1.3 Power-Slip-Characteristic (Type 2 only)

Type 2 needs further configuration of the power dependent slip characteristic, this characteristic can be entered on the *Load Flow* page of the asynchronous generator.

3.1.4 Model Aggregation

The WTG can also be used for aggregated representation of several WTGs or even a whole wind farm. The only setting needed for this is the “Number of parallel machines” on the *Basic Data*

page of the element. The active power output of the generator is then the active power dispatch times the number of parallel machines. The set-up transformer has to be adjust accordingly.

3.2 Type 3 and 4A/B

3.2.1 Adjusting Rated Power

The rated power of the static generator has to be configured on the *Basic Data* page of the element itself (i.e. the static generator doesn't need a type). The rated active power is calculated via the nominal apparent power "sgn" and the power factor "cosn".

3.2.2 Adjusting Power Dispatch

The dispatch of the generator can be entered on the *Basic Data* page. The default bus type is "PQ". Using this bus type allows to enter the active and reactive power dispatch. The reactive power can be limited by the reactive power capability curve which can be also voltage dependent. It is also possible to coordinate the reactive power output of several static generators by using a station controller (*ElmStactrl*).

3.2.3 Model Aggregation

The WTG can also be used for aggregated representation of several WTGs or even a whole wind farm. The only setting needed for this is the "Number of parallel machines" on the *Basic Data* page of the element. The active power output of the generator is then the active power dispatch times the number of parallel machines. The set-up transformer has to be adjust accordingly.

3.3 Dynamic Model Configuration

The dynamic simulation is started from steady state, which is the load flow result. It is therefore important to check the load flow configuration of the elements. The configuration for Type 1 and 2 is described in chapter 4.1, and chapter 4.2 describes Type 3 and 4 configuration.

The dynamic model can either be accessed via the generator element in the single line diagram or via the data manager.

- Via Element: Open the element. Go to the *Basic Data* page and click on the arrow button after "Plant Model" (Type 1 and 2) or at "Model" (Type 3 and 4).
- Via Data Manager: Search the corresponding composite model , right click and select "Edit".

After opening the composite model the different dynamic (common) models can be accessed via a double click in the 2nd column.

Note: The models which are available in *PowerFactory* as templates are configured with a set of sample parameters. It is task of the user to adjust these parameters (i.e. power rating, voltage level, reactive power limits and all parameters of the dynamic models) to represent

a specific WTG of a specific manufacturer!

The dynamic (common) models can be configured via scalar input parameters. These parameters are displayed after opening a common model on the *General* page.

Some models, such as the “WT1P_B Pitch Control Model” (Type 1 and 2), the “WT1P_B Pitch Control Model” (Type 2), the “WTGTRQ_A Torque Control Model” (Type 3) and the “REEC_D Electrical Control Model” (Type 3, 4A and 4B) contain also arrays (i.e. a list of parameters). The arrays can be found on the second page of the common model *Advanced 1*. Each array consists of two columns. The dimension can be changed by modifying the number in the top left cell of an array. The minimum dimension is two. The page has to be refreshed after changing the dimension by switching once to the “General” page and then back to “Advanced 1”.

4 Model Documentation

In the following sub-sections the different types are described. In Table 4.1 the different dynamic models are listed.

Table 4.1: Block Overview

DSL Model	Block Definition	1	2	3	4A	4B
WT12T Drive-Train Model	WT12T	x	x			
WT1P_B Pitch Control Model	WT1P_B	x	x			
WT2E External Resistance	WT2E		x			
WTGTRQ_A Torque Control Model	WTGTRQ_A			x		
WTGPT_A Pitch Control Model	WTGPT_A			x		
WTGAR_A Aero-Dyn Model	WTGAR_A			x		
WTGT_A Drive-Train Model	WTGT_A			x	x	
WTGIBFFR_A Inertial-Base Fast-Frequency	WTGIBFFR_A			x	x	x
WTGWGO_A Weak Grid Option Model	WTGWGO_A				x	x
REEC_D Electrical Control Model	REEC_D			x	x	x
REGC_C Generator-Converter Model	REGC_C			x	x	x
Protection Model	Protection	x	x	x	x	x
Plant Control	Frame WECC Plant Control			(x)	(x)	(x)

4.1 Type 1 and 2

Type 1 and 2 WTG are represented in the network via the asynchronous generator element and a shunt capacitor which is used to compensate the reactive power demand of the generator. The initial settings for the load flow calculation and dynamic simulation are done in the asynchronous machine. The configuration of the grid element is described in 3.1.

Type 1 and 2 WTG represent a conventional squirrel-cage induction generator and a wound-rotor induction generator with variable rotor-resistance respectively, the models as build in *PowerFactory* consist of the following components:

- Generator.
- WT12T, which represents the standard two-mass drive train model.
- WT1P_B, which implements the active power ramp down and up after a voltage dip.
- WT2E, which implements the external resistance controller, together with the active power ramp (down and up controller).
- Protection, which implements the protection functions with disconnection from the grid in case of over- or under-frequency, over- or under-voltage.
- Voltage and power measurement.

4.2 Type 3 and 4

Type 3 and 4 WTG are represented in the network via the static generator element. The initial settings for the load flow calculation and dynamic simulation are done in the static generator, configuration of the grid element is described in section 3.2.

The generic Type 3 and Type 4A/B WTG as build in *PowerFactory* consist of the following components:

- Generator. This represents the WTG in the network.
- WTGT_A, which implements the behaviour of the wind turbine generator torsional mode oscillations.
- WTGAR_A, which implements a simple linear model of the wind turbine generator aerodynamics.
- WTGPT_A, which implements a simplified representation of the pitch controller.
- WTGTRQ_A, which implements two possible operation modes based on speed or power error to compute the power command.
- WTGIBFFR_A, which implements the auxiliary inertial-based fast-frequency response, initiated in case of under-frequency events and system frequency decrease [8].
- WTGWGO_A, which implements a supplemental weak grid option control [8].
- REEC_D, which implements the electrical control model. The main task is the control of the reactive power. As shown in Table 4.2, different control models are available.
- REGC_C, which implements the fast converter controllers and represents the interface between the generator/converter [9].

- Protection, which implements the protection functions with disconnection from the grid in case of over- or under- frequency, over- or under- voltage.
- Plant Control, which implements the plant controller in charge of determining the commanded active and reactive current. The plant control is available as a template in the templates folder “Plant Controllers”, see also chapter 5.
- Voltage, power and current measurement.

Table 4.2: Type 3 and Type 4A/B electrical control modes [2]

Control Mode	**PfFlag**	**Vflag**	**Qflag**
Local constant Q control	0	1	0
Local constant pf control	1	1	0
Local voltage control	0	0	1
Local coordinated Q/V control	0	1	1

Note: The REGC_C model is by default applied in the Type 3 and Type 4A/B templates; however, the REGC_A or REGC_B model can be used instead if desired [4]. Both, REGC_A model and REGC_B model type, are available within the DIgSILENT Library\Dynamic Models\Inverter Based Resources/Storage\WECC.
The REGC_B and REGC_C model are defined as a controlled voltage source [9]. The main difference between the REGC_B and REGC_C is that the last one includes a simplified phase-lock loop (PLL) model. The former REGC_A model was defined as a controlled current source [1].
The static generator element is configured to be a *voltage source* model in RMS simulation, i.e. to represent a controlled voltage source in accordance with REGC_C or and REGC_B.
If the REGC_A model is used, the *current source* model has to be selected in the RMS simulation settings of the static generator element. *DIgSILENT* recommends to use the additional built-in option *dq-reference angle delay* of the static generator for better numerical robustness, especially when simulating deep voltage dips. This option is described in detail in the technical reference paper of the static generator element [3]. The *delay time constant* should correspond to a typical time constant of a PLL.

4.3 Measurement

Power, voltage and current measured signals are implemented in the models using the *PowerFactory* inbuilt elements *StaPqmea*, *StaVmea* and *StaImea* correspondingly [6], [7], [5].

In all models, the frequency signal needed in the Protection Model is taken from the voltage measurement device *StaVmea*, please read the technical reference paper of this device to learn more about the frequency calculation.

5 Plant Controller

In addition to the four WT template types a plant level controller template as specified in [1] is also provided. The “WECC Plant Controller 50Hz” and “WECC Plant Controller 60Hz” template is accessible using the button in the drawing toolbox.

The plant controller allows to control either voltage, with droop or line drop compensation, the reactive power or the powerfactor (last one only available in REPC_C). Additionally an active power control function including a simple droop-down function in case of over-frequencies is also included, as optional feature an up regulation function is incorporated too. A dedicated flag can be set to enable or disable active power control.

The composite model of the plant control (“Plant Control Model”) has to be linked into the slot “Plant Control” of the WTG to connect the models and the power rating of the power measurement (one or two depending on the plant control model used) have to be adjusted to the plant rated power. The plant control model is intended to be used with aggregated plant representation only. It can be used in principle also in a more detailed wind farm representation but this requires that all WTG units are initialized with the same active and same reactive power because the “Plant Control” will send to all WTG units the same signal.

Note: The REPC_C model is by default applied in the Plant Control model template; however, the REPC_A model can be used instead if desired. The REPC_A model type is available within the DIgSILENT Library/Dynamic Models.
The REPC_C model includes a coordinated switching of shunt capacitors and reactors functionality [9], which can be enable or disable. Within the model up to ten shunt devices can be included.

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6 References

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[7] *Technical Reference "Voltage Measurement Device"*. DIgSILENT GmbH, Gomaringen, Germany, 2023.

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[9] P. Pourbeik. Proposal for new features for the renewable energy system generic models 07/23/18 (revised 8/3/18; 11/14/18; 11/18/18; 12/13/18; 1/20/19; 2/7/19; 3/5/19; 6/24/19; 6/28/19; 7/3/19; 7/17/19; 8/19/19; 12/17/19; 8/14/20; 8/24/20;8/31/20; 11/12/20; 4/6/21; 4/15/21; 9/22/21; 11/11/21; 9/19/22; 12/12/22; 1/5/23). 2023.

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